Idovonzas Torvenye - Time-Attraction Law
Unified Effective Density Framework - Idokamra 1.0 v8.3 FINAL

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Main DOI: 10.5281/zenodo.23138388 | Note DOI: 10.5281/zenodo.22899435 | Concept DOI: 10.5281/zenodo.22983682
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Abstract (EN)

This document provides the final corrected derivation for Time-Attraction Law v8.3. Corrigendum v8.2 fixes R=(r+1)/(r+2) to the correct Bellini & Sawicki (2014) eq 2.18 R=(1+r)/(1+2r) where r=alpha_M/alpha_B. Update v8.3 derives epsilon in closed form on subclass G5=0, G4=G4(phi): M*^2=2G4, alpha_M=2 phidot G4phi/(H M*^2), H M*^2 alpha_B=2 phidot [X G3x - G4phi] -> r=G4phi/(X G3x-G4phi), epsilon=X G3x/G4phi -2. Consequently f_L cancels: r=1/(1+epsilon), low-density limit independent of screening form. Sign sorted via dR/dr=-1/(1+2r)^2<0. Background upgrade replaces local rho_L with background rho_m(a)=rho_m0 a^-3, so f_L->f_L(a), alpha_M(a), alpha_B(a) are time functions as required by EFT of dark energy, yielding binned R(a) testable via hi_class against Euclid/DESI. No hand-chosen 1.2 and 2, shape falsifiable. SPARC rotation curves contain no per-galaxy lensing, thus DeltaSigma not available and R cannot be formed from SPARC; 76-95% accuracy framing removed, replaced by hi_class binned R(a) forecast (template Sec X enhanced-early-gravity cubic Galileon recovers Planck mass within 0.03% across 4 redshifts, 3 wavenumbers). Framework remains falsifiable, dimensionally consistent (kg/m3). Acknowledgment to Jordi Audet Palau for Sec V, VI, X corrections.

1. Corrigendum v8.2 - R formula javitasa

```
v8.1-ben hibasan R=(r+1)/(r+2) szerepelt. Helyes forma Bellini & Sawicki (2014) Eq 2.18 es
Jordi Audet Palau note alapjan:
R = DeltaSigma/DeltaMu = (1+r)/(1+2r), r = alpha_M/alpha_B
Inverz: r = (1-R)/(2R-1)
Ellenorzes: r=1 -> R=2/3 mindket kepletben, de r!=1 eseten elter: r=0.5: regi 0.60 -> helyes
0.75; r=2: regi 0.75 -> helyes 0.60; r=5: regi 0.857 -> helyes 0.545
dR/dr = -1/(1+2r)^2 < 0 monoton csokkeno.
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2. Update v8.3 - epsilon zart alak levezetese (Per Jordi Sec V)

```
Feltetel: Horndeski alosztaly G5=0, G4=G4(phi) -> G4X=0. EFT parameterek:
M*^2 = 2 G4
alpha_M = 2 phidot G4phi / (H M*^2)
H M*^2 alpha_B = 2 phidot [X G3X - G4phi]
Osztas: r = alpha_M/alpha_B = G4phi / (X G3X - G4phi)
1+epsilon = alpha_B/alpha_M = 1/r -> epsilon = 1/r -1 = X G3X/G4phi -2 QED
Locus: epsilon=0 <-> X G3X = 2 G4phi <-> r=1 <-> R=2/3
f_L kiesik: r = alpha0 f_L / [alpha0 f_L (1+epsilon)] = 1/(1+epsilon) -> fuggetlen screening
formatol
Elojell: dR/dr<0 -> X G3X >2 G4phi -> epsilon>0 -> r<1 -> R felulrol; X G3X <2 G4phi ->
epsilon<0 -> r>1 -> R alulrol (Sec4 ezt keri).
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3. Background Correction v8.3

```
Problema: alpha_M, alpha_B csak idofuggo hatterfuggvenyek EFT-ben. Lokalis rho_L nem
parameteruk -> f_L(rho_L) nem illesztheto.
Megoldas: rho_L legyen hatter: rho_m(a)=rho_m0 a^-3 -> f_L->f_L(a), alpha_M(a), alpha_B(a),
epsilon(a)=X G3X/G4phi -2, r(a)=1/(1+epsilon(a)), R(a)=(1+r(a))/(1+2r(a)) binned a-ban ->
hi_class -> Euclid/DESI teszt.
Template Sec X: enhanced-early-gravity cubic Galileon r=-1.2 to -1.3 visszahozza Planck
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tomeget 0.03%-on.
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4. Sec5 SPARC Revision

SPARC = rotacios gorbek + Spitzer fotometria, nincs per-galaxis lensing -> DeltaSigma nincs -> R nem kepezheto SPARC-bol. 76-95% framing torolve. Csere: hi_class binned R(a) forecast Sec VI. Regi 94.9% SPARC 150 -> Uj R(a) binned Euclid/DESI.

5. Unified Framework kg/m3

```
rho_eff = rho_b*M_rho + rho_DM*M_rho + rho_B*M_B + rho_GR*M_B + rho_E*M_B + rho_psi
rho_psi = Xi*epsilon0*Phi_t, Phi_t=0.6*EEG_coh+0.4*HRV_coh
v8.3: rho_L->rho_m(a), epsilon(a), r(a), R(a) - dimenzio-helyes, falszifikalhato.
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6. Kepletek egy helyen - Fuggelek

```
R = (1+r)/(1+2r)
r = (1-R)/(2R-1) = G4phi/(X G3X - G4phi) = 1/(1+epsilon)
epsilon = X G3X/G4phi -2
dR/dr = -1/(1+2r)^2
M*^2=2G4, alpha_M=2 phidot G4phi/(H M*^2), H M*^2 alpha_B=2 phidot[X G3X-G4phi]
rho_m(a)=rho_m0 a^-3, R(a)=(1+r(a))/(1+2r(a)) binned
Phi_t=0.6*EEG_coh+0.4*HRV_coh
```